

Chapter 9: System Design

Learning Objectives

By the end of this chapter, you will:

- Understand the principles of designing scalable, secure, and reliable AI systems.
 - Learn common enterprise architecture patterns used for AI applications.
 - Develop the ability to participate in System Design discussions as an AI TPM.
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Beginner Explanation

System Design is the process of planning how an application will work before it is built.

Imagine building a shopping mall. Before construction begins, you need to answer questions such as:

- How many people will visit daily?
- Where will parking be located?
- What happens during power failures?
- How will security be managed?
- How will the mall expand in the future?

AI systems require the same planning.

Questions an AI TPM Should Ask

Question	Example
Scalability	Can this support 100,000 users?
Availability	What happens if Azure OpenAI is unavailable?
Security	How is sensitive data protected?
Performance	What is the acceptable response time?
Cost	How much will the solution cost monthly?

Core System Design Principles

1. Scalability
2. Reliability
3. Availability
4. Security
5. Observability
6. Maintainability

Good system design ensures the application continues to work even when components fail.

Intermediate Explanation

Enterprise AI systems typically include multiple components working together.

Typical AI Architecture

Layer	Purpose
Front End	User interaction
API Layer	Request management
AI Services	Reasoning and generation
Search Layer	Knowledge retrieval
Data Layer	Storage and persistence
Monitoring Layer	Observability
Security Layer	Access management

Non-Functional Requirements (NFRs)

Requirement	Target Example
Availability	99.9%
Latency	<3 Seconds
Throughput	10,000 Requests/Hour
Recovery Time Objective (RTO)	<30 Minutes
Recovery Point Objective (RPO)	<15 Minutes

System Design Trade-Offs

Decision	Benefit	Trade-Off
Larger Model	Better quality	Higher cost
Caching	Lower latency	More complexity
Multi-Region	High availability	Increased cost
Serverless	Easier operations	Cold starts

An AI TPM should understand these trade-offs to support architectural decisions.

Why This Matters for an AI TPM

Responsibility	Impact
Delivery	Ensures scalable and reliable solutions are delivered.
Risk	Identifies performance, security, and availability risks early.
Stakeholder Management	Communicates technical decisions to business stakeholders.
Budget	Balances architecture decisions with cost constraints.

Enterprise Example

Business Problem

A global bank wanted to deploy an AI-powered customer assistant expected to handle more than 2 million requests per month.

Key requirements included:

- 99.9% uptime
- Multi-region deployment
- Regulatory compliance
- Less than 3-second response time

Solution

Build a highly available Azure-based AI architecture with:

- Azure OpenAI
- Azure AI Search
- Azure API Management
- Azure Functions
- Azure Front Door
- Azure Monitor

Outcome

Metric	Result
Availability	99.95%
Average Latency	2.4 Seconds
Monthly Requests	2.3 Million
Customer Satisfaction	94%

Lessons Learned

- Monitoring should be implemented from Day 1.

- Multi-region deployments significantly improve reliability.
 - Security reviews are mandatory.
 - Caching reduces operational costs.
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Azure Implementation

Services Used

Service	Purpose
Azure OpenAI	AI inference
Azure AI Search	Enterprise retrieval
Azure Functions	Workflow orchestration
Azure AI Foundry	Evaluations and prompt management

Additional Services

- Azure Front Door
 - Azure API Management
 - Azure Monitor
 - Azure Key Vault
 - Azure Blob Storage
 - Azure Application Insights
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Implementation Workflow

Step 1: Define Non-Functional Requirements

Examples:

- Availability: 99.9%
 - Latency: <3 seconds
 - Scalability: 100K users
 - Security: RBAC
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Step 2: Design Architecture

Determine:

- Components
 - Integrations
 - Dependencies
 - Failure scenarios
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Step 3: Implement Security

Include:

- Private Endpoints
 - RBAC
 - Key Vault
 - Network Isolation
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Step 4: Enable Observability

Monitor:

- Response time
 - Errors
 - Costs
 - User activity
 - Availability
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Step 5: Conduct Load Testing

Validate:

- Peak traffic handling
 - Failure recovery
 - Performance under load
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Mermaid Diagram

```
```mermaid id="cyulh1" graph TD
```

```
A[Users]
```

```
A --> B[Azure Front Door]
```

```
B --> C[API Management]
```

```
C --> D[Azure OpenAI]
```

```
D --> E[Azure AI Search]
```

```
E --> F[Azure Functions]
```

```
F --> G[Blob Storage]
```

```
G --> H[Azure Monitor]
```

```

High Availability Architecture

```mermaid id="0myqov"
graph LR

A[Region 1]
--> B[Azure OpenAI]

A --> C[Azure AI Search]

D[Region 2]
--> E[Azure OpenAI]

D --> F[Azure AI Search]

B --> G[Traffic Manager]

E --> G

G --> H[Users]

```

TPM Checklist

- ☐ Define NFRs.
- ☐ Conduct architecture reviews.
- ☐ Validate scalability requirements.
- ☐ Define disaster recovery plans.
- ☐ Conduct security assessments.
- ☐ Perform load testing.
- ☐ Establish monitoring.
- ☐ Review monthly costs.
- ☐ Create runbooks.
- ☐ Obtain architecture approvals.

Templates

Architecture Review Checklist

Item	Status
Scalability Reviewed	Yes/No
Security Approved	Yes/No
DR Plan Created	Yes/No

Item	Status
Monitoring Enabled	Yes/No
Costs Estimated	Yes/No

Non-Functional Requirements Template

Requirement	Target
Availability	99.9%
Latency	<3 Seconds
Throughput	10K Requests/Hour
RTO	<30 Minutes
RPO	<15 Minutes

Risk Register

Risk	Impact	Mitigation
Service Failure	High	Multi-region deployment
High Costs	Medium	Budget alerts
Security Incident	High	Security reviews
Performance Issues	Medium	Load testing

Common Mistakes

1. Ignoring non-functional requirements.
2. Designing for today's traffic instead of future growth.
3. Skipping disaster recovery planning.
4. Underestimating security requirements.
5. Neglecting monitoring and observability.
6. Failing to perform load testing.
7. Making architecture decisions without cost analysis.

Key Takeaways

- System Design is a foundational skill for every AI TPM.
- Enterprise AI solutions must be scalable, secure, and highly available.
- Non-functional requirements are as important as functional requirements.
- Azure provides services for building resilient AI architectures.

- Monitoring, security, and disaster recovery should be considered from the beginning.
- AI TPMs must understand architectural trade-offs to effectively lead enterprise programs.

Footer

charan.ai | Page 43